

YEAR 10 Earth and Space Sciences

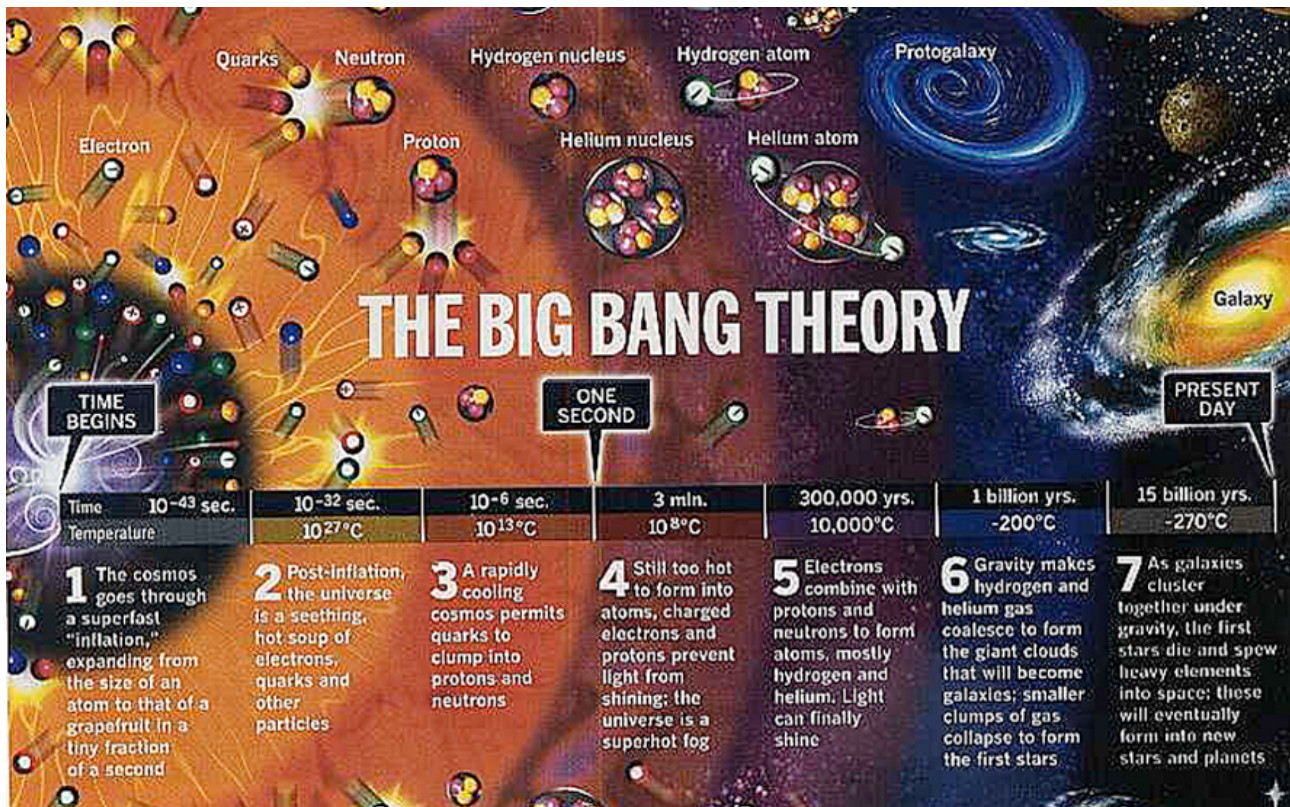
Title	The Big Bang Theory - Research report			
Learning area	Science			
Year Level(s)	10			
Brief Description of Assessment Activity	In this task students are required to undertake research that enables them to outline the origin of the universe as described by the Big Bang Theory. They are also asked to discuss the contributions of scientists in the development of the theory and elaborate on the involvement of Australian scientists. Students present their research in the form of a written report. There is no word limit specified. Students are encouraged to provide a reference list of the sources used to gather their information.			
Context summary	Students have spent time in class studying the origins of the Universe and the enormous distances involved in the overall expanse of the universe. Whilst working on this report they are also looking more closely at the origins of the Universe and the various theories that have been proposed over time			
Purpose	Formative		Summative	
Description of purpose	Assessing knowledge of the Big Bang Theory and how and why this has become the favourite theory. Also their ability to construct evidence based arguments and communicate their ideas. (see achievement Standard below)			
Audience Suitability	At Year Level	Extension	Students with disability	EAL/D

Links to the Australian Curriculum

Content Strand	Science Understanding		Science as a Human Endeavour			Science Inquiry Skills		
Sub-strand	Biological Sciences		Use and influence of science			Questioning and predicting		
	Chemical Sciences					Planning and conducting		
	Earth and Space Sciences		Nature and development of science			Processing and analysing data and information		
	Physical Sciences					Evaluating		
						Communicating		
General capabilities	Literacy	Numeracy	ICT capability	Critical and creative thinking	Ethical behaviour	Personal and social competence	Intercultural understanding	
Cross-curriculum priorities	Aboriginal and Torres Strait Islander histories and cultures			Asia and Australia's engagement with Asia			Sustainability	

Content descriptions	Relevant Aspects of the Achievement Standard
Strand Substrand - The universe contains features including galaxies, stars and solar systems and the Big Bang theory can be used to explain the origin the universe (ACSSU188) - Scientific understanding, including models and theories, are contestable and are refined over time through a process of review by the scientific community (ACSHE191) - Use knowledge of scientific concepts to draw conclusions that are consistent with evidence (ACSI204) - Communicate scientific ideas and information for a particular purpose, including constructing evidence-based arguments and using appropriate scientific language, conventions and representations (ACSI208)	By the end of Year 10, students analyse how the periodic table organises elements and use it to make predictions about the properties of elements. They explain how chemical reactions are used to produce particular products and how different factors influence the rate of reactions. They explain the concept of energy conservation and represent energy transfer and transformation within systems. They apply relationships between force, mass and acceleration to predict changes in the motion of objects. Students describe and analyse interactions and cycles within and between Earth's spheres. They evaluate the evidence for scientific theories that explain the origin of the universe and the diversity of life on Earth. They explain the processes that underpin heredity and evolution. Students analyse how the models and theories they use have developed over time and discuss the factors that prompted their review. Students develop questions and hypotheses and independently design and improve appropriate methods of investigation, including field work and laboratory experimentation. They explain how they have considered reliability, safety, fairness and ethical actions in their methods and identify where digital technologies can be used to enhance the quality of data. When analysing data, selecting evidence and developing and justifying conclusions, they identify alternative explanations for findings and explain any sources of uncertainty. Students evaluate the validity and reliability of claims made in secondary sources with reference to currently held scientific views, the quality of the methodology and the evidence cited. They construct evidence-based arguments and select appropriate representations and text types to communicate science ideas for specific purposes.

WHERE DID WE COME FROM?????



The research that you need to do for this assignment is divided into two main parts. You can either decide to deal with these as separate parts of your discussion or you can mention both aspects throughout your discussion. You **MUST** provide us with

- ALL your research notes
- Your written report
- Your reference list

Your research report shall

- give an explanation of the Big Bang Theory as it stands today and show how the theory came about.
- give the names and some history of the important people who helped to develop the theory. Are there any Australians who have played a part?

CHECK THE RUBRIC TO MAKE SURE YOU HAVE HIT EACH OF THE CRITERIA WE HAVE INCLUDED FOR MARKING!!!!

(Hint: Einstein and Stephen Hawking are two scientists I would expect to see mentioned in your research - mind you, they are NOT Australian or the only ones!!!!)

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Achievement Standard (Science)	ND	NW	C	HC	O
Students evaluate the evidence for scientific theories that explain the origin of the universe.					
I can give a clear explanation for the Big Bang Theory of the origin of the universe					
I can present a range of evidence that led to the development of the Big Bang theory					
I can clearly explain how the evidence supports the Big Bang Theory					
Students analyse how the models and theories they use have developed over time and discuss the factors that prompted their review.					
I can explain how theories are developed					
I can explain how new evidence can change or make scientists review a theory (related to the Big Bang Theory)					
I have acknowledged the discovery of recent evidence by an Australian scientist that prompted further review and modification of the Big Bang Theory.					
They construct evidence-based arguments and select appropriate representations and text types to communicate science ideas for specific purposes					
I have been able to link evidence to any modifications to the theory each time a new piece of evidence has been proposed					
I have used words appropriate to a general audience					
I have referenced ALL my sources of information					
I have made all information that I have given easy to read and understand by making sure I have explained all scientific terms that I have used.					